

MGMT 298D: THE SCIENCE AND STRATEGY OF ARTIFICIAL INTELLIGENCE

Spring 2026 – 4 Units – In-Person

Course Syllabus

REVISED 11/30/2025

Teaching Team Information

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Course meeting times and location

<i>Course Day & Time:</i>	[Day and time]
<i>Course Location:</i>	[Room number]
<i>Course Site:</i>	[BruinLearn link]

Course Description

Catalog Description

Introduces essential artificial intelligence (AI) models and their role in shaping strategy across industries. Students gain hands-on experience with AI models in Python with practical use cases. Culminates in a team project prototyping an AI-powered micro-app published on the web.

Expanded Description

This course aims to build students' familiarity with essential AI models, and to deepen understanding of how they are deployed in practice. Designed for a non-technical audience, the course demystifies core AI concepts, unpacking them into accessible and intuitive frameworks. The course emphasizes experiential learning through guided, hands-on sessions in Python. No prior coding experienced is needed.

Students will learn how to frame strategic challenges as prediction problems, understand what AI can (and cannot) do, and deploy AI models that solve real-world business problems. Core topics include supervised, unsupervised, and reinforcement learning, neural networks, image recognition, natural language processing,

and AI agents. Business use cases will cover demand forecasting, pricing, customer segmentation, personalized recommendations, risk management, and summarization.

In the final two-week module, students will team up to build an AI-powered micro-app to be published on the web. Drawing on their learnings from the quarter, each team will identify a specific business need addressed by AI and prototype a simple app, showcasing their skills to potential employers and serving as a memento of the course.

Course Objectives

At the end of the course, students will be able to:

- 1. Frame business problems as prediction tasks and evaluate the suitability of AI solutions**
Understand how AI and machine learning models can be used to drive strategic decisions in domains such as marketing, operations, and product design.
- 2. Implement and apply core AI methods**
Understand the mathematical and computational foundations of AI and gain hands-on experience with Python-based AI tools (e.g., Google Colab, sklearn, Keras) to build AI models for uses cases spanning demand forecasting, pricing, customer segmentation, personalization, image recognition, and AI agents.
- 3. Design and communicate AI-powered business solutions**
Collaborate in teams to conceptualize, prototype, and present an AI-powered micro-app that addresses a specific business need, integrating technical know-how with strategic thinking.

Learning Tools

We will make use of Google Colab's free notebook environment, which will enable us to implement various AI models using Python code. We will rely on some of the most well-known Python packages for data science and deep learning: sklearn, pandas, matplotlib, and Keras. We will also make use of LLMs such as Google Gemini to assist with coding. The micro-app project in the final weeks of the course will be published online using the streamlit package in Python.

Course Materials

Main textbook:

Deep Learning with Python, Francois Chollet (2021)

Optional texts and media:

[Book] *The Hundred-Page Machine Learning Book*, Andriy Burkov (2019)

[Book] *The AI Playbook: Mastering the Rare Art of Machine Learning Deployment*, Eric Siegel (2024)

[Video] *Deep Dive into LLMs like ChatGPT*, Andrej Karpathy (2025)

<https://www.youtube.com/watch?v=7xTGNNLPyMI>

[Video] *The spelled-out intro to neural networks*, Andrej Karpathy (2022)

<https://www.youtube.com/watch?v=VMj-3S1tku0>

[Website] 3Blue1Brown: *Neural Networks* (2017).

<https://www.3blue1brown.com/topics/neural-networks>

[Book] *Neural Networks and Deep Learning*, Michael Nielsen (2019)

<http://neuralnetworksanddeeplearning.com/>

[Articles] Distill <https://distill.pub/> (Plain English explanations of AI research)

Course Outline

Week	Date	Title and Description	Readings	Deliverable
1	DD/MM/YY	Linear Regression with Feature Engineering <i>Use case: Retail Demand Forecasting</i> Building intuition for model-fitting, using a case study on retail demand forecasting with H&M transaction data. We introduce core ideas for predictive modeling in high dimensions, including linear regression, LASSO regularization, train/test splits, under- and overfitting, and cross-validation. Students will build a demand forecasting model and explore how feature engineering improves accuracy using the sklearn package in Python, within Google Colab.		A1 (Ungraded)
2	DD/MM/YY	Tree-Based Predictions <i>Use case: Algorithmic Pricing</i> This module focuses on AI-driven pricing, using Carvana as a case study to illustrate the strategic differentiation enabled by algorithmic pricing. We cover classification and regression trees, random forests, and XGBoost. Students will build vehicle pricing models and compare performance across algorithms using the sklearn package in Python.	Chapter 6, <i>Deep Learning with Python</i> (Chollet, 2021) Algorithms Need Managers, Too (HBR, 2016)	A2
3	DD/MM/YY	Clustering <i>Use case: Consumer Segmentation</i> Introduction of unsupervised learning methods motivated by consumer segmentation, using Netflix ratings data as a case study in how firms group users to support personalization. Students learn clustering ideas—distance metrics, the k-means algorithm, and cluster evaluation—and apply k-means to Netflix data identify viewer segments. Students will conduct the analysis in Python using sklearn and standard visualization tools.	Darden Case M-0748: Cluster Analysis for Segmentation Visualizing K-Means Clustering https://www.naftaliharris.com/blog/visualizing-k-means-clustering/	A3
4	DD/MM/YY	Reinforcement Learning <i>Use case: Personalized Recommendations</i> This module focuses on dynamic decision-making, motivated by robotics, dynamic pricing, and recommendations. We introduce core ideas in reinforcement learning, including state–action spaces, rewards, and the exploration–exploitation trade-off. Students will train a Q-learning agent using the gymnasium package in Python and observe how decisions evolve with experience.	A Revolution in How Robots Learn (The New Yorker, 2024) Amazon made \$1 billion through secret price raising algorithm -US FTC, Reuters (2023) The big new idea for making self-driving cars that can go anywhere (MIT Technology Review, 2023)	A4

Week	Date	Title and Description	Readings	Deliverable
5	DD/MM/YY	Neural Networks <i>Use case: Volatility Forecasting and Risk Management</i> Introduction of neural networks, using a case study in volatility forecasting for risk management using S&P 500 data. We cover neural network architecture, including neurons, activation functions, hidden layers, and backpropagation, and compare NNs with linear and tree-based models. Students will build neural networks using the Keras package in Python to forecast volatility and experiment with hyperparameters such as depth, units, learning rate, and lookback window.	Chapters 2, 3 & 4, <i>Deep Learning with Python</i> (Chollet, 2021) <i>How Deep Learning Works</i> , IEEE Spectrum, September 2021 <i>AI Is Helping Pick What You'll Wear in Two Years</i> , Bloomberg, January 31, 2024	A5
6	DD/MM/YY	Computer Vision <i>Use case: Object Detection and Autonomous Vehicles</i> Overview of computer vision applications in autonomous vehicles, medical imaging, and e-commerce. Students learn the basics of convolutional neural networks and how CNNs differ from dense networks. Students will train a CNN using the Keras package in Python to classify different images and see a short in-class demo on object detection.	Chapters 7 & 8, <i>Deep Learning with Python</i> (Chollet, 2021) <i>Foundation Models for Generalist Medical Artificial Intelligence</i> , Nature, 2023 <i>The Limits of Computer Vision, and of Our Own</i> , Harvard Medicine Magazine, Spring 2024 <i>What Self-Driving Cars Tell Us About AI Risks</i> , IEEE Spectrum, 2023	A6
7	DD/MM/YY	Transformers & LLMs <i>Use case: Document Summarization</i> Overview of the transformer architecture that has powered the AI revolution of the 2020s. We cover tokenization, embeddings, and self-attention, and explain how transformers enable modern LLMs. Students will train a small transformer in Keras using a corpus of thousands of Reuters articles.	Chapter 11, <i>Deep Learning with Python</i> (Chollet, 2021) <i>Generative AI Exists Because of the Transformer. This Is How It Works</i> (Financial Times, 2023) <i>What Is Generative AI?</i> (IEEE Spectrum, 2024)	A7
8	DD/MM/YY	Generative AI <i>Use case: AI Shopping Agent</i> Focus on generative and agentic AI, using product-review analysis as a case study. Students learn how generative models produce text, based on concepts such as temperature and top-k sampling, and discuss how LLMs can operate as simple agents. Students will use the Google Gemini API to build a mock “shopping agent” that analyzes Amazon reviews and recommends a product.	Chapter 12, <i>Deep Learning with Python</i> (Chollet, 2021) <i>Generative AI Holds Much Promise for Businesses</i> , (The Economist, 2023) <i>A Short History of Tractors in English: What the Tractor and the Horse Tell You About Generative AI</i> (The Economist, 2023)	A8

Week	Date	Title and Description	Readings	Deliverable
9	DD/MM/YY	Prototyping and Publishing <i>Use case: You Decide!</i> Turning machine-learning and AI ideas into lightweight applications. Students learn rapid prototyping and deployment concepts, then work in teams to build a small ML-powered web app. Students will implement and publish their app on the web using the streamlit package in Python.	Streamlit documentation: https://docs.streamlit.io/	Product Sketch
10	DD/MM/YY	Product Pitches Effective communication of AI-driven solutions. Teams refine their apps and deliver short product pitches that articulate business value, demonstrate functionality, and explain key technical elements. Students will present their deployed Streamlit applications to peers and the instructor.		Group Presentation

Evaluation and Grading

This course will be graded using the following weighted percentages for each of the components in the course. Feedback and grades are typically posted within one week of assignment due dates.

Assessment	% of Grade
Assignments (10% each) [Individual]	70%
Micro-App Prototype [Groups]	25%
Peer Evaluation [Individual]	5%
Total	100%

Your overall course grade will be determined by how your performance on graded assignments ranks in comparison with other students in the class according to the grade distribution model at Anderson. Note that courses in which an overall grade of C is received must be offset by higher grades in the same term for students to remain in good academic standing at UCLA. A grade of C- or below in a core class must be retaken.

Description of Assessments

Assignments (70%)

There are seven assignments, each worth 10%, designed to help students apply course concepts in practical settings using data and modeling tools. Each assignment will consist of submitted a Python notebook completed in and downloaded from Google Colab.

Assignments will be submitted individually but collaboration is heavily encouraged. The course will include in-class time to complete the assignments, though students may need to finalize their work outside of class. Each assignment builds on technical and strategic skills relevant to AI model development, such as model building, evaluation, and interpretation.

Micro-App Prototype (25%)

The final project challenges students to prototype an AI-powered micro-app. Working in teams, students will identify a business use case, develop a working model or concept, and deliver a final pitch that communicates the business value, technical feasibility, and deployment strategy. The project emphasizes the integration of business insight, technical design, and communication skills. The micro-app will be published on the web using the Streamlit package in Python.

Peer Evaluation (5%)

To promote a collaborative environment, each student will evaluate their teammates' engagement and contributions to labs and the final project. These evaluations are fully confidential.

Anderson and Course Policies

AI Usage Policy

This course will allow the use of AI (Large Language Models, including ChatGPT, Copilot, Gemini, and others) to enhance your learning, not replace it! The goal is to expand student capabilities and output in research and analysis, providing greater time to contemplate, analyze, and develop more sophisticated analyses, recommendations, and leadership imperatives. All students have access to Microsoft Copilot, the AI-powered chat mode of Microsoft using GPT 4 Turbo, a secure and protected tool once you login with your UCLA ID.

To facilitate the achievement of this goal, every assignment in the class will state whether using AI is appropriate. If you are confused about what is allowed, please ask the instructor or the course assistant for clarification. When using AI for individual or group assignments, you must acknowledge the tool's name and how it was used using the following style.

Netiquette

The written language has many advantages: more opportunity for reasoned thought, more ability to go in-depth, and more time to think through an issue before posting a comment. However, written communication also has certain disadvantages, such a lack of the face-to-face signaling that occurs through body language, intonation, pausing, facial expressions, and gestures. As a result, please be aware of the possibility of miscommunication and compose your comments in a positive, supportive, and constructive manner.

UCLA Policies

Code of Conduct

All participants in the course are bound by the **UCLA Student Conduct Code** (<https://deanofstudents.ucla.edu/individual-student-code>) and **UCLA Anderson Honor Code** (<https://www.anderson.ucla.edu/documents/areas/adm/web/AndersonHonorCode.pdf>).

Academic Integrity

UCLA is an institution of learning, research, and scholarship predicated on the existence of an environment of honesty and integrity. As members of the academic community, instructors, students, and administrative officials are all responsible for maintaining this environment. It is essential that all members of the academic community practice academic honesty and integrity and accept individual responsibility for their work. Academic misconduct is unacceptable and will not be tolerated in this course. Cheating, forgery, dishonest conduct, plagiarism, and collusion in academic misconduct erode the University's educational, research, and social roles.

Students who knowingly or intentionally conduct or help another student engage in acts that violate UCLA's expectations of academic integrity will be subject to disciplinary action and referred to the Dean of Students' Office.

Please familiarize yourself with **UCLA's Academic Integrity Policy**:

<https://www.deanofstudents.ucla.edu/Academic-Integrity>. Speak to your instructor if you have any questions about what is and is not allowed in this course.

Integrity in Research

Integrity in research includes not just the avoidance of wrongdoing, but also the rigor, carefulness, and accountability that are hallmarks of good scholarship. All persons engaged in research at the University are responsible for adhering to the highest standards of intellectual honesty and integrity in research.

Please familiarize yourself with the **University of California Policy on Integrity in Research**

(<https://www.ucop.edu/academic-personnel-programs/files/apm/apm-190-b.pdf>)

Accessible Education & Inclusive Education

Disability Services

UCLA is committed to providing a barrier-free environment for persons with documented disabilities. If you are already registered with the Center for Accessible Education (CAE), please request your Letter of Accommodation in the Student Portal. If you are seeking registration with the CAE, please submit your request for accommodation via the CAE website. Students with disabilities requiring academic accommodations should submit their request for accommodations as soon as possible, as it may take up to two weeks to review the request. For more information, please visit the CAE website (www.cae.ucla.edu), visit the CAE at A255 Murphy Hall, contact CAE by phone at (310) 825-1501, or by telecommunication device for the deaf at (310) 206-6083.

Equity, Diversity, and Inclusion

Please familiarize yourself with UCLA Anderson's commitment to maintaining an equitable, diverse, and inclusive community:

(<https://www.anderson.ucla.edu/about/equity-diversity-and-inclusion>)

Appendix A: Comparable Courses at US MBA Programs

The following is a non-exhaustive list of MBA electives in the area of AI and machine learning offered at U.S. business schools. Most of the courses below have a “hands-on” component with respect to developing AI models, similar to the proposed course.

Columbia Business School

Foundations of AI for Business

<https://courses.business.columbia.edu/B8143>

Harvard Business School

Data Science for Managers

<https://dsm.business/intro.html>

Johns Hopkins Carey Business School

Data Science: Artificial Intelligence

<https://tinglongdai.com/teaching/>

MIT Sloan School of Management

The Analytics Edge

<https://ocw.mit.edu/courses/15-071-the-analytics-edge-spring-2017/>

Hands-on Deep Learning

<https://www.coursicle.com/mit/courses/15/773/>

Northwestern Kellogg School of Management

AI Foundations for Managers

<https://www6.kellogg.northwestern.edu/CourseCatalog/coursecatalog/coursedetail?coursecatalogid=208012>

NYU Stern School of Business

Machine Learning for Business

https://w3.stern.nyu.edu/syllabi/static/syllabusfiles/TECH-UB.57_001_S2025.pdf

Stanford Graduate School of Business

Data Science and AI Strategy

<https://www.aistanford.org>

OIT 367 Business Intelligence from Big Data

<https://web.stanford.edu/~bayati/oit367.html>

Wharton School, University of Pennsylvania

Applied Machine Learning in Business

<https://catalog.upenn.edu/search/?P=STAT%204230>

Yale School of Management

Generative AI and Social Media

<https://coursetable.com/catalog?course-modal=202501-22421>

Appendix B: Response to FEC Curriculum Subcommittee Comments

FEC Curriculum Subcommittee: *Recognized as strategically valuable for deepening technical AI skills derived from foundational coursework. However, given the course's anticipated launch in 2027 and current lack of detailed syllabus, the committee expects a more thorough syllabus be provided closer to the offering date, clearly specifying course structure, content depth, and distinctions from other AI-focused courses. The 2-1 vote indicates some reservations about the current syllabus needing to be more fully fleshed out, but we are approving this because we deep faith in the proposals thoughtfulness in addressing an urgent need for students capabilities in this domain.*

I am thankful for the helpful comments from the FEC Curriculum Subcommittee. The syllabus now contains a week-by-week description of specific topics, relevant readings, and technological aids (e.g., software packages). The description of each week is now more detailed to outline the key learning objectives and business use cases of the AI models discussed. The descriptions of assessments have also been revised.

Regarding the distinction from other AI courses, this course will be the only AI-focused MBA course at Anderson that fully immerses students in working with data, code, and models. The final two-week sprint to develop an AI-powered micro-app deployed on the web is also novel. This syllabus was shaped by consulting with other instructors at Anderson and is also based on discussions from the AI Curriculum working group, on which I served. For reference, I include below the course comparison table from the memo generated by that committee.

Course Name	Primary Focus	Coding?	Best For...	Hands-On Highlights
MGMT 275:AI Essentials	Core concepts & tools for fluency	Some	All functions(General Mgmt)	Prompt design, productivity tools, in-class experiments
MGMT 275:Leading with AI	Applying AI to Operations	Some	Ops & Product(Cross-functional)	Use-case development, org integration exercises
MGMT 276:Leading in the AI Era	Strategic & ethical leadership	Some	Future Leaders(GMs, Strategy)	Decision frameworks, AI policy, leadership case studies
MGMT 289D:Science & Strategy	Machine learning & Deep learning	Yes	Tech Strategy(Consulting)	ML pipelines, deep-learning models, micro-app development
MGMT 298E:AI Agents	Building autonomous agents	Some	Builders(Innovation roles)	Agent workflows, chaining, building prototypes
MGMT 298D:AI & Real Estate	Analytics for Real Estate	Some	Real Estate(PropTech)	Industry chatbots, valuation models, RE tech cases
Capstones(AMR, GAP, SMR, MP)	Live client challenges	Some	Project-Based(All tracks)	Client deliverables, agent testing, AI prototypes